Status	Finished
Started	Thursday, 8 October 2026, 12:28 PM
Completed	Thursday, 8 October 2026, 12:32 PM
Duration	4 mins 27 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There should be direct access to the middle element in any sublist

b. Number values should only be present

Number values should only be present

c. The list must be sorted

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **2**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n

b. n ✓

c. n-1

d. n/2

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)

b. O(log n)

c. O(n log n)

d. O👎 ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Binary search
c. Jump search
d. Linear search ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Linear search ✓
b. Quick search
c. Merge search
d. Bubble search

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 90 and 94
c. 89 and 99
d. 90 and 99 ✓

Question **7**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search works on unsorted lists
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. At the beginning of the list
b. In the right sublist
c. In the left sublist ✓
d. In the entire list

Question **9**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Compare the target element with the middle element in the list ✓
c. Sort the list
d. Divide the list into two equal parts

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 500
c. 100
d. 1000

Question **11**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. log n
c. n ✓
d. n/2

Question **12**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the list is sorted
- c. When the list contains duplicate elements
- d. When the element is found in the list ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Binary search ✓
- b. Linear search
- c. Sequential search
- d. Hash search

Question **14**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(n log n)
- b. O(log n)
- c. O👎
- d. O(1) ✓

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It does not work with negative numbers
- b. It is slower than linear search for small lists
- c. It requires the list to be sorted ✓
- d. It can only be applied to large lists